

JAYANT KUSHWAHA

M.Sc. BIOTECHNOLOGY

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MSc Biotechnology student, dissertation at ICMR-NICPR on cervical cancer genomics. My published work (Virus Genes, 2025) looked at inhibiting Monkeypox virus proteins through computational screening PPI networks, molecular docking, enrichment analysis. On the wet-lab side, I've done DNA extraction, ELISA, and SDS-PAGE. I want to keep working at the crossover between genomics and computation, ideally in cancer biology or infectious disease research.

EDUCATION

MASTERS OF SCIENCE IN BIOTECHNOLOGY	2024 - 2026
• NOIDA INTERNATIONAL UNIVERSITY	C.G.P.A – 9.10
▪ Dissertation: Systems-level Analysis Of Cervical Cancer Related Dysregulated Genes Using Computational Approaches	
BACHELORS OF SCIENCE IN BIOTECHNOLOGY	2021 - 2024
• R.N.C Arts J.D.B Commerce N.S.C Science College Nashik.	C.G.P.A – 8.66
▪ Dissertation: Synthesizing Zinc oxide Nanoparticles from Banana peels and its applications.	

EXPERIENCE

MASTER'S DISSERTATION	Jan – 2026 to April - 2026
ICMR - NATIONAL INSTITUTE OF CANCER PREVENTION AND RESEARCH.	
• Curated literature-derived gene expression data from cervical cancer studies and compiled a dysregulated gene set for downstream computational analysis.	
• Ran functional enrichment via Enrichr, mapping the gene set to oncogenic pathways.	
• Built protein-protein interaction networks in STRING and visualized them in Cytoscape to identify central network nodes based on degree and betweenness centrality.	
• AKT1 and EGFR emerged as hub genes both are well-established in cervical cancer biology, which adds confidence to the network model.	
INTERNSHIP	23rd Jan – 28th Jan 2023
AGRI SEARCH (INDIA) PVT LTD.	
• Acquired knowledge about manufacturing process carried out in Agri Search and obtained understanding of water assessment.	
• Studied Quality Control and Quality Assurance techniques required in industries.	
• Experienced how the fermentation processes is done.	
• Observed how the Bio-fertilizers are prepared and tested.	

SKILLS

Laboratory Techniques: Gel Electrophoresis, DNA extraction, SDS-PAGE, Plant Tissue Culturing, ELISA, Protein Estimation, PCR

Bioinformatics Tools:- NCBI, BLAST, GenBank, PDB, Molecular Docking, PyMOL, Discovery Studio, Enrichr, Cytoscape, MD Simulations.

Programming Skills:- C/C++, Bio Python.

PROJECTS

- **Structure-Based Virtual Screening of Inhibitors Against Monkeypox Virus Proteins**
Independent Research Project – 2025
 - Modeled MPXV proteins using AlphaFold2.
 - Retrieved and prepared ligands from PubChem for docking.
 - Performed virtual screening using AutoDock Vina.
 - Analyzed protein-ligand interactions with PyMOL and Discovery Studio.
- **Synthesis of Zinc oxide Nanoparticles and its Application**
Undergraduate Dissertation Project -2024
 - Synthesized zinc oxide nanoparticles from Banana peels extract by bottom-up approach.
 - Characterization by UV-Vis Spectrophotometer, FTIR, SEM.
 - Tested the Antimicrobial-activity assay on *E.coli* and *S.aureus*.
 - Applied on Mung plants and studied their chlorophyll and Protein content.

PEER-REVIEWED PUBLICATIONS

- Kumari, R., **Kushwaha, J. M.**, Sharma, V. K., Shrivastav, D., & Singh, S. (2026). Distinct molecular pathways driving lung cancer in non-smokers: Etiology, signaling networks, and clinical implications. *Critical Reviews in Oncology/Hematology*, 226, 105436.
<https://doi.org/10.1016/j.critrevonc.2026.105436> **I.F. 6.2**
- **Kushwaha, J.M.**, Rani, M.S.S. & Singh, S. Targeting monkeypox virus (MPXV): strategies for molecular docking studies on protein inhibition. *Virus Genes* (2025).
<https://doi.org/10.1007/s11262-025-02171-1> **I.F. 1.9**

BOOK CHAPTERS

- Singh S.*, **Kushwaha, J.M.** (2026). "Next-Generation Biotechnology for Jaivik Kheti: Biofertilizers & Plant Growth Promoters." In *Organic Agriculture and Sustainable Farming Systems* (Eds. Hasan F., Singh A.), pp. 15-19. Learning Media Publication, Meerut. ISBN 978-81-69199-14-8.

CERTIFICATES

ACADEMIC:

- **Focus on Peer Review, Nature Masterclasses – July 2025**
 - Completed 3.5 hours of expert-led training on peer review fundamentals. Covered ethical reviewing, critical manuscript evaluation, and reviewer responsibilities. Enhanced ability to provide constructive feedback and improve scientific writing.
- **ELISA Workshop – March 2025**
 - Completed two-days hands-on workshop on Advanced Biological Techniques ELISA and did the quantitative estimation of TSH in Human plasma sample.
- **Bioinformatics Internship'25 –February 2025**
 - Completed the 5 days internship which focused on Bioinformatics tools, SNP Data Analysis and Building AI Genomics Prediction model.
- **C/C++ Programming Language –August 2023**
 - Completed the Certificate Programming language course on C/C++ at **Disha Computers**.

FORAGE:

- **LifeArc Life Sciences: Biology Research Job Simulation on Forage - January 2025**
 - Completed a job simulation involving optimizing experimental conditions for LifeArc's Biology Research team.
 - Analyzed the differentiation protocol for optimization needs.
- **Tata Data Visualization: Empowering Business with Effective Insights Job Simulation on Forage - August 2024**
 - Completed a simulation involving creating data visualizations for Tata Consultancy Services.
 - Prepared questions for a meeting with client senior leadership.
 - Created visuals for data analysis to help executives with effective decision making.

PROFILES



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[WOS ID- NVM-0932-2025](#)



[Scopus ID- 59959633300](#)



[Research Gate ID- Jayant-Kushwaha](#)

PERSONAL DETAILS

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| • Date of Birth: 14/10/2003. | • Gender: Male |
| • Marital Status: Single. | • Nationality: Indian |
| • Languages Known: English, Hindi, Marathi. | |